By contrast (as we saw in Chapter 17), on the orbit of Miller's planet, the balance between the gravitational and centrifugal forces is stable.

Disaster on the Rim: Ejection of TARS and Cooper

In my science interpretation of the movie, the volcano's rim is very narrow, so the critical orbit on the rim is exceedingly unstable. Tiny errors in navigation will send the *Endurance* careening down toward Gargantua (down into the volcano) or away from Gargantua (down toward the moat).

Errors are inevitable, so the *Endurance*'s course must be corrected, continually, by a well-designed feedback system, like an automobile's cruise control but much better.

In my interpretation, the feedback system is not quite good enough and the *Endurance* winds up dangerously far down the inside lip of the volcano. The *Endurance* must use all the thrust at its disposal to climb back up to the critical orbit.

But this is too subtle and technical for action-packed scenes and a hugely diverse audience, so Christopher Nolan chose a simpler, more in-your-face approach. No mention of instability. No mention of feedback. The *Endurance* simply plunges too close to Gargantua, and Cooper responds with all the thrust he can muster to climb back out and escape Gargantua's grip.

The result is the same: lander 1, piloted by TARS, and Ranger 2, piloted by Cooper, fire their rockets while attached to the *Endurance*, pushing the *Endurance* back out of Gargantua's gravitational grip. Then, to get the last possible kick, explosive bolts blow the *Endurance* apart from lander 1 and Ranger 2. The lander and Ranger go plunging